

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Second-Semester Examination, May, 2017

First (1st) Year

Agric, Aerospace, Biomedical, Chemical, Mechanical, Material and Petroleum Engineering

COE/EE 152: BASIC ELECTRONICS

TIME: 2 HRS

PROGRAM OF STUDY: MATERIALS ENGINEERING

INDEX NO: [illegible]

Instructions

i. Candidates are to indicate their Index Number and Program of Study in the spaces provided.
ii. Candidates are to submit ONLY the Scannable Sheet.
iii. Choose from the options lettered A to D, the most suitable answer to the question statements
iv. Shade the most suitable answer to the question statements on the Scannable Sheet provided
v. Candidates are to write the answer at the back of the Scannable Sheet if he/she thinks the right answer cannot be found in the options provided
vi. Each correct answer carries half (½) a mark
vii. 50 questions are online on the virtual classroom from 8th to 12th May 2017

JOSEPH ATSU TAMAKLOE wishes you a fruitful semester Rising Heights #CONSISTENCY #SPEED #GRACE ATSU '18 GESA PRO '18 BELLWADE | Art Tomorrow, Today

"BELIEVE IN THE PROCESS" ATSU FOR GESA P.R.O

1. If the transistor is not biased properly, it would work …….. and produce distortion in output signal
    a. Inefficiently
    b. Efficiently
    c. As a switch
    d. None of the above
2. A transistor is biased either with the help of ……
    a. Battery
    b. Associating a circuit
    c. Both of the above
    d. None of the above
3. Which of the following influences changes in the collector current of a transistor
    a. Temperature
    b. Transistor parameters
    c. Both of the above
    d. None of the above
4. The maintenance of operating point stable is called ……
    a. Amplification
    b. Transformation
    c. Stabilisation
    d. Switching Circuit
5. The selection of a proper quiescent point generally depends on the following factors:
    a. The amplitude of the signal to be handled by the amplifier and distortion level in signal
    b. The load to which the amplifier is to work for a corresponding supply voltage
    c. Both of the above
    d. None of the above
6. The operating point of a transistor amplifier shifts mainly with changes in……,
    a. Temperature
    b. Humidity
    c. Fog
    d. Mist
7. ……., …… and …….are functions of temperature
    a. β, I_{CO} and V_{EE}
    b. β, I_{CO} and V_{CC}
    c. β, I_{CO} and V_{CE}
    d. β, I_{CO} and V_{BE}
8. For a transistor circuit to amplify it must be properly biased with……..
    a. ac currents
    b. ac voltages
    c. dc voltages
    d. dc currents

"BELIEVE IN THE PROCESS" ATSU FOR GESA P.R.O

55. Which of the following will be useful in producing an n-type semiconductor?
    a. Acceptor atoms
    b. Donor atoms
    c. Pentavalent impurity
    d. Silicon
56. To operate properly, a transistor's base-emitter junction must be forward biased with reverse bias applied to which junction?
    a. collector-emitter
    b. base-collector
    c. base-emitter
    d. collector-base
57. A *pn* junction allows current flow when
    a. the p-type material is more positive than the n-type material
    b. the n-type material is more positive than the p-type material
    c. both the n-type and p-type materials have the same potential
    d. there is no potential on the n-type or p-type materials
58. For a given BJT circuit, the ends of a load line drawn on a family of curves determine:
    a. saturation and cutoff
    b. the operating point
    c. the power curve
    d. the amplification factor
59. In an *npn* transistor, the majority carriers in the emitter are
    a. Free electrons
    b. Holes
    c. Neither
    d. Both
60. Most of the electrons in the base of an *npn* transistor do not recombine because they
    a. Have a long lifetime
    b. Have a negative charge
    c. Must flow through the base
    d. Flow out of the base
61. What denotes the maximum voltage a diode can withstand when reverse-biased?
    a. Peak inverse voltage
    b. Peak forward-bias voltage
    c. Reverse breakdown
    d. Reverse bias
62. In the active region, the base-emitter junction _____.
    a. and the base-collector junctions are both forward-biased
    b. and the base-collector junctions are both reverse-biased
    c. is reverse-biased while the base-collector junction is forward-biased
    d. is forward-biased while the base-collector junction is reversed-biased

63. Why is design for a specific bias point desirable for most amplifiers?
    a. To meet manufacturer suggested opening point.
    b. It allows optimum dc operation of the circuit.
    c. It allows optimum ac operation of the circuit.
    d. All of the above
64. A given transistor has ratings of maximum collector current equal to 200 mA and a beta that varies between 150 and 200.What is the maximum allowable value of base current for the device?
    a. 1 mA
    b. 4 mA
    c. 1.33 mA
    d. None of the above
65. Which amplifier configuration is the least often used?
    a. Common Base
    b. Common Collector
    c. Darlington Pair
    d. Common Emitter
66. When a Zener diode circuit is used to stabilize the output voltage given a fixed load resistor and a variable input voltage, the input voltage must be _____.
    a. large enough to turn off the Zener diode
    b. small enough to turn on the Zener diode
    c. small enough to turn off the Zener diode
    d. large enough to turn on the Zener diode
67. Transistor circuits that are quite stable and relatively insensitive to temperature variations have_____.
    a. relative high supply voltages
    b. small betas
    c. large betas
    d. low supply voltages
68. Given that a BJT has V_{CE} of 13 V and V_{CC} is 15 V, how would you describe this BJT.
    a. properly biased
    b. approaching cutoff
    c. a pnp transistor
    d. approaching saturation
69. The majority carriers in the p region of a diode
    a. a much fewer in number than the majority carriers in the n region
    b. are electrons
    c. a much less in number than the majority carriers in the n region
    d. are holes
70. Doping is used to _____.
    a. increase the conductivity of an intrinsic semiconductor
    b. increase the insulative quality of an intrinsic semiconductor
    c. stabilize the conductivity of an intrinsic semiconductor
    d. decrease the conductivity of an intrinsic semiconductor

71. The term bipolar means:
    a. the use of both holes and electrons as carriers
    b. electrons make up the majority carriers
    c. polarity is unimportant
    d. diodes only
72. The maximum reverse bias potential that can be applied to a Zener diode before it enters the Zener region is called the _____.
    a. threshold voltage
    b. PIV
    c. depletion voltage
    d. barrier voltage
73. When a p-n junction is reverse-biased, its junction resistance is _____.
    a. Low
    b. high
    c. constantly changing
    d. determined by the components that are external to the device
74. What can be said regarding the energy level of an electron?
    a. Electrons exhibit energy characteristics only in the presence of protons
    b. Electrons in the shell closest to the nucleus of an atom have the highest energy level
    c. Electrons in the outermost shell have the highest energy level
    d. All electrons have the same energy level
75. Carriers are injected..... into the base from the emitter
    a. transmitted
    b. transformed
    c. emitted
    d. transferred
76. Since the base of a typical transistor is thin, most carriers injected into the base .....into the.....
    a. diffuse, emitter
    b. diffuse, collector
    c. infuse, emitter
    d. infuse, collector
77. Base thickness of a typical BJT is much thinner than the .....of the carrier injected from the.....
    a. diffusion length, emitter
    b. infusion strength, collector
    c. diffusion length, collector
    d. infusion length, emitter
78. The arrow on a BJT symbol indicates the direction of the ..... current.
    a. collector
    b. emitter
    c. base
    d. transistor

In biasing a PNP transistor one p-n junction of a transistor is reverse-biased, whereas the other is forward-biased in the common base configuration. Draw a diagram of the PNP transistor and use the diagram to answer **questions 79 and 80**

79. The large number of .....will diffuse across the reverse-biased junction into the p-type material connected to the collector terminal.



    a. electrodes
    b. majority carriers
    c. ions
    d. minority carriers

80. .....can cross the reverse-biased junction because the injected majority carriers will appear as ..... in the n-type material.
    a. majority carriers, minority carriers
    b. minority carriers, majority carriers
    c. minority carriers, diffusion
    d. diffusion, minority carriers

81. Which of the following is not true about the active mode of operation of a BJT?
    a. Most important mode of operation
    b. Central to amplifier operation
    c. The region where current curves are practically flat
    d. Current is reduced to zero.

82. In the .....the barrier potential of the junctions of a BJT cancel each other out causing a virtual short.
    a. cutoff region
    b. saturation region
    c. active region
    d. inverse active region

83. In the .....the current passing through the BJT is reduced to zero and the transistor behaves ideally like an open switch.
    a. cutoff region
    b. saturation region
    c. active region
    d. inverse active region

84. The region where the current curve of a BJT is practically flat is referred to as.....
    a. cutoff region
    b. saturation region
    c. active region
    d. inverse active region

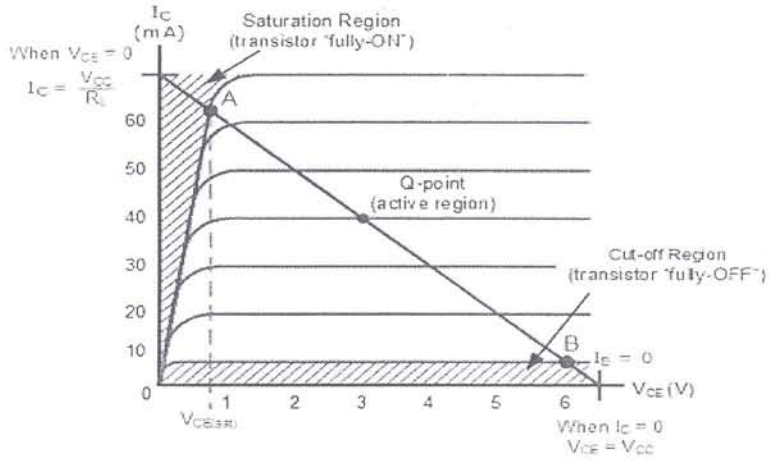


Figure 1.11

Use figure 1.11 to answer questions 85 to 90

85. For small values of V_{CE} ($V_{CE} < V_{CE(SAT)}$), what is the relationship between I_C and V_{CE}.
    a. Constant
    b. Exponential
    c. Linear
    d. Quadratic
86. When $V_{CE} > V_{CE(SAT)}$ I_C is not..... on V_{CE} and I_C is practically.....
    a. dependant, continuous
    b. dependant, changing
    c. dependant, varying
    d. totally dependent, constant
87. I_B is usually in.....and ..... compared to I_C which is in.....
    a. nA and large, MA
    b. μA and small, mA
    c. kA and small, μA
    d. μA and large, kA
88. A small increase in I_B causes a(n) .....increase in I_C
    a. small
    b. large
    c. medium
    d. insignificant
89. When $I_B = 0$ μA .....occurs.
    a. I_{CBO}
    b. I_{CEO}
    c. I_{CO}
    d. h_{FE}
90. When I_C is zero,.....
    a. the transistor is short circuited
    b. the transistor is open circuited
    c. there is still some values of current flowing
    d. all of the above

9. The dc operating point between saturation and cutoff is called the .....
    a. Action point
    b. Q-point.
    c. Amplification point
    d. None of the above
10. The goal is to set the .....such that it does not go into saturation or cutoff when an AC signal is applied.
    a. Turn-on value
    b. Q-point
    c. Temperature
    d. Amplification
11. Which of the following is **NOT** a requirement of biasing networks/circuits
    a. Ensuring proper zero signal collector current.
    b. Ensuring V_{CE} not falling below 0.5V for Ge transistor and 1V for Silicon transistor at any instant.
    c. Ensuring Stabilization of operating point. (zero signal I_C and V_{CE})
    d. None of the above
12. The DC current gain (β) of a transistor remains constant at 100 at 25°C and the base current (I_B) at 20.28μA while the collector current (I_C) changes from 2.028mA to 3.04mA, evaluate the thermal stability factor (S) of the operating point of this transistor.
    a. 1
    b. 25
    c. 50
    d. None of the above
13. Which of the following is a condition for the solution in **question 12** to remain always true?
    a. I_B and alpha must remain constant
    b. I_B and alpha must vary
    c. I_B and beta must remain constant
    d. I_B and beta must vary

$\alpha = \frac{I_C}{I_E}$

14. Which of the following is the desired range of values for the thermal stability factor (S) of the operating point of a transistor?
    a. 1 to 50
    b. 1 to 25
    c. 1 to 75
    d. Has no range
15. Which of the following is the ideal value(s) for the thermal stability factor (S) of the operating point of a transistor?
    a. 1
    b. 25
    c. 50
    d. 75

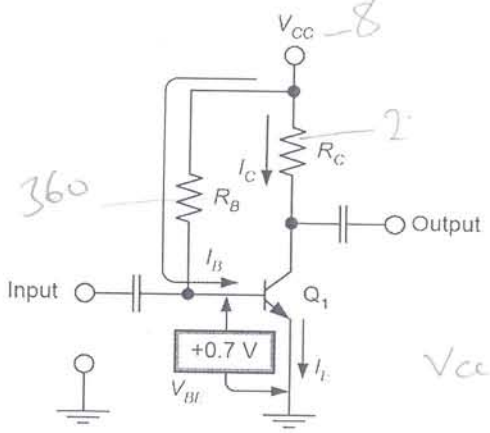


Figure 1.0

The transistor in Fig. 1.0 has values of $h_{FE} = 100$ when T = 25 °C and $h_{FE} = 150$ when T = 100 °C. Given that $R_B = 360k\Omega$, $R_C = 2k\Omega$, and $V_{CC} = +8V$

Use figure 1.0 and the information below it to answer questions from 16 to 21

16. Determine the value(s) of the base current at both of these temperatures
    a. 20.28μA
    b. 2.028mA
    c. 20.28mA
    d. 2.028μA
17. Determine the Q-point value(s) of I_C and V_{CE} at both of these temperatures.
    a. 20.28 μA and 4.93V
    b. 2.028 mA and 3.94V
    c. 20.28 mA and 4.93V
    d. 2.028 μA and 3.94V
18. Determine the value(s) of I_C and V_{CE} at Saturation and Cut-off respectively.
    a. 4.028 μA and 4.00V
    b. 4.280 mA and 6.93V
    c. 4.000 mA and 8.00V
    d. 2.028 μA and 7.94V
19. Which of the following is an advantage and a disadvantage of the biasing technique employed in figure 1.0
    a. Circuit simplicity and Q-point shift with change in temperature
    b. Good Amplification and Q-point shift with change in temperature
    c. High Stability and Q-point shift with change in temperature
    d. None of the above
20. The values of I_C and V_{CE} at T = 25 °C compared to the values of I_C and V_{CE} at Saturation and Cut-off respectively it can be said that.....
    a. Average bias circuit
    b. Mid-point bias circuit
    c. Maximum bias circuit
    d. None of the above

21. Which of the following is a(n) application for the biasing type shown in figure 1.0
    a. Switching
    b. Trimming
    c. Regulating
    d. Amplification

The diodes used in a bridge rectifier circuit have the following parameters:
$V_T = 0V$, $r_f = 10\Omega$, $R_R = \infty$, V_Z = very large
The secondary winding of the transformer has a resistance $R_S = 10\Omega$. A load of 970Ω is connected across its output. If the voltage across the secondary is $v_s = 20 \sin wt$ volts,

Use the above information to solve questions 22 to 29

22. Determine the PIV rating of the diodes
    a. 40V
    b. 20V
    c. 60V
    d. 10V
23. Determine the DC current through the load
    a. 0.160A
    b. 0.128A
    c. 0.78408A
    d. 0.20406A
24. Determine the DC power supplied to the load
    a. 0.160W
    b. 0.128W
    c. 0.78408W
    d. 0.20406W
25. What is the conversion efficiency of this rectifier
    a. 0.160pu
    b. 0.128pu
    c. 0.78408pu
    d. 0.20406pu
26. What is the peak value of the AC current?
    a. 0.2020A
    b. 0.2128A
    c. 0.78408A
    d. 0.20406A
27. What is the AC power supplied to the whole circuit?
    a. 0.2020W
    b. 0.2128W
    c. 0.78408W
    d. 0.20406W

28. What is the RMS value of the current supplied to the circuit?
    a. 0.1428A
    b. 0.1248A
    c. 0.2428A
    d. 0.20406A
29. Which of the following will improve the efficiency of the rectifier
    a. Employing a capacitor for filtering with practically zero internal resistance
    b. Employing a zener diode for regulation with ideal internal resistance
    c. Employing a diode with practically zero internal resistance
    d. Employing a transformer with practically zero internal resistance

Preamble for questions (30 to 33). A load of 3W is connected across a 30V zener regulator. The zener current can vary from 50mA to 100mA whilst maintaining a constant voltage output.

30. Find the minimum input voltage if the series resistor is 1kΩ.
    a. 150V
    b. 200V
    c. 250V
    d. 300V
31. Find the maximum input voltage if the series resistor is 1kΩ.
    a. 150V
    b. 200V
    c. 250V
    d. 300V
32. What is the maximum power the zener diode can withstand?
    a. 4W
    b. 3W
    c. 3.5W
    d. 3.9W
33. How would you determine if the zener diode used in the circuit is ON or OFF?
    a. Comparing the voltage ratings of the zener diode to the voltage drop across the load.
    b. Comparing the voltage ratings of the zener diode to the voltage drop across the filter capacitor.
    c. Comparing the voltage ratings of the zener diode to the voltage drop across the zener diode.
    d. Comparing the voltage ratings of the zener diode to the voltage drop across the transformer.

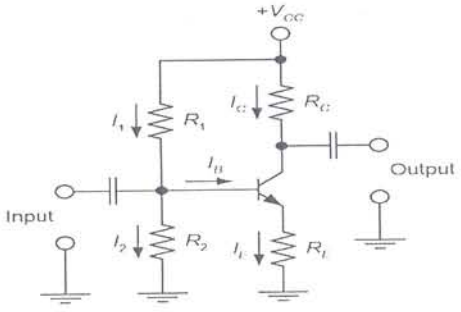


Figure 1.10

The transistor in Fig. 1.10 has $h_{FE} = 50$ when at room temperature, T = 25 °C. Given that $R_1 = 18k\Omega$, $R_2 = 4.7k\Omega$, $R_E = 1.1k\Omega$, $R_C = 3k\Omega$, and $V_{CC} = +10V$.

Use figure 1.10 and the information below it to answer questions from 34 to 42

34. Determine the base voltage (V_B) on the transistor assuming the base current (I_B) is negligible.
    a. 2.12V
    b. 2.17V
    c. 2.21V
    d. 2.207V

35. Determine the actual value of the base current I_B
    a. 2.451μA
    b. 2.451mA
    c. 24.51μA
    d. 24.51mA

36. Determine the value of I_2
    a. 44.04μA
    b. 440.4μA
    c. 4.404μA
    d. 4404.0μA

37. Which of the following is a TRUE relationship between I_2 and I_B?
    a. $I_2 << I_B$
    b. $I_2 << 10\ I_B$
    c. $I_2 >> I_B$
    d. $I_2 >> 10\ I_B$

38. Given that the relationship you selected in **question 37** is TRUE, what will be the actual value of the voltage on the base of the transistor.
    a. 1.27V
    b. 2.07V
    c. 2.70V
    d. 3.207V

39. Determine the value of the collector current (I_C) at the Q-point.
    a. 125mA
    b. 12.5mA
    c. 1.25mA
    d. 0.125mA

40. Determine the value of the collector emitter voltage (V_{CE}) at the Q-point.
    a. 7.48V
    b. 4.87V
    c. 8.47V
    d. 3.487V
41. What is the value of (I_C) at saturation?
    a. 3.44mA
    b. 3.44μA
    c. 2.44μA
    d. 2.44mA
42. What is the value of (V_{CE}) at cut-off?
    a. 20.0V
    b. 9.98V
    c. 9.99V
    d. 10.0V

Assuming that the relationship you selected in **question 37** is **NOT TRUE** and the transistor in Figure. 1.10 has $h_{FE} = 50$ when at room temperature, T = 25 °C. Given that $R_1 = 68k\Omega$, $R_2 = 10k\Omega$, $R_E = 1.1k\Omega$, $R_C = 6.2k\Omega$, and $V_{CC} = +20V$.

Use figure 1.10 and the above information to answer questions from 43 to 48

43. What will be the actual value of the voltage on the base of the transistor?
    a. 2.12V
    b. 2.07V
    c. 2.21V
    d. 2.207V
44. Determine the value of the collector current (I_C) at the Q-point.
    a. 1.37mA
    b. 1.37μA
    c. 2.73μA
    d. 2.73mA
45. Determine the value of the collector emitter voltage (V_{CE}) at the Q-point.
    a. 20.0V
    b. 9.98V
    c. 9.99V
    d. 10.0V
46. What is the value of (I_C) at saturation?
    a. 1.37mA
    b. 1.37μA
    c. 2.73μA
    d. 2.73mA
47. What is the value of (V_{CE}) at cut-off?
    a. 20.0V
    b. 9.98V
    c. 9.99V
    d. 10.0V

48. Which of the following can be considered as a(n) disadvantage of the biasing type shown in figure 1.10
    a. Requires more capacitors than most other biasing circuits
    b. Requires more components than most all other biasing circuits
    c. Requires more components than most fixed bias circuits
    d. Requires more components than most emitter bias circuits

Preamble for questions (49 to 51) Given that a transistor has a $\beta = 50$ and $I_E = 5.00mA$

49. What is the value of I_B?
    a. 0.98μA
    b. 9.8μA
    c. 98μA
    d. 0.098μA

$\frac{5}{(50+1)} = I_B = 0.098$

50. What is the value of I_C?
    a. 5.98mA
    b. 5.098mA
    c. 598.0mA
    d. 59.8mA

$I_B = I_E + I_C$

$I_C =$

51. Find the value of the ratio I_B: I_C.
    a. 1:21
    b. 1:32
    c. 1:42
    d. 1:52

52. The amount of time between the creation of a hole and its disappearance in a semiconductor material is called
    a. Doping
    b. Lifetime
    c. Recombination
    d. Valence

53. When a voltage is applied to a semiconductor, holes will flow
    a. Away from the negative potential
    b. Toward the positive potential
    c. In the external circuit
    d. None of the above

54. While maintaining a constant temperature, a silicon diode has its reverse-bias voltage increased. The diode's saturation current will
    a. Increase
    b. Decrease
    c. Remain the same
    d. Equal its surface-leakage current